Integrative Industry Synthesis

Agentic Fraud Detection via Dual-Threshold ML Classification and Generative Data Augmentation & Benchmarking

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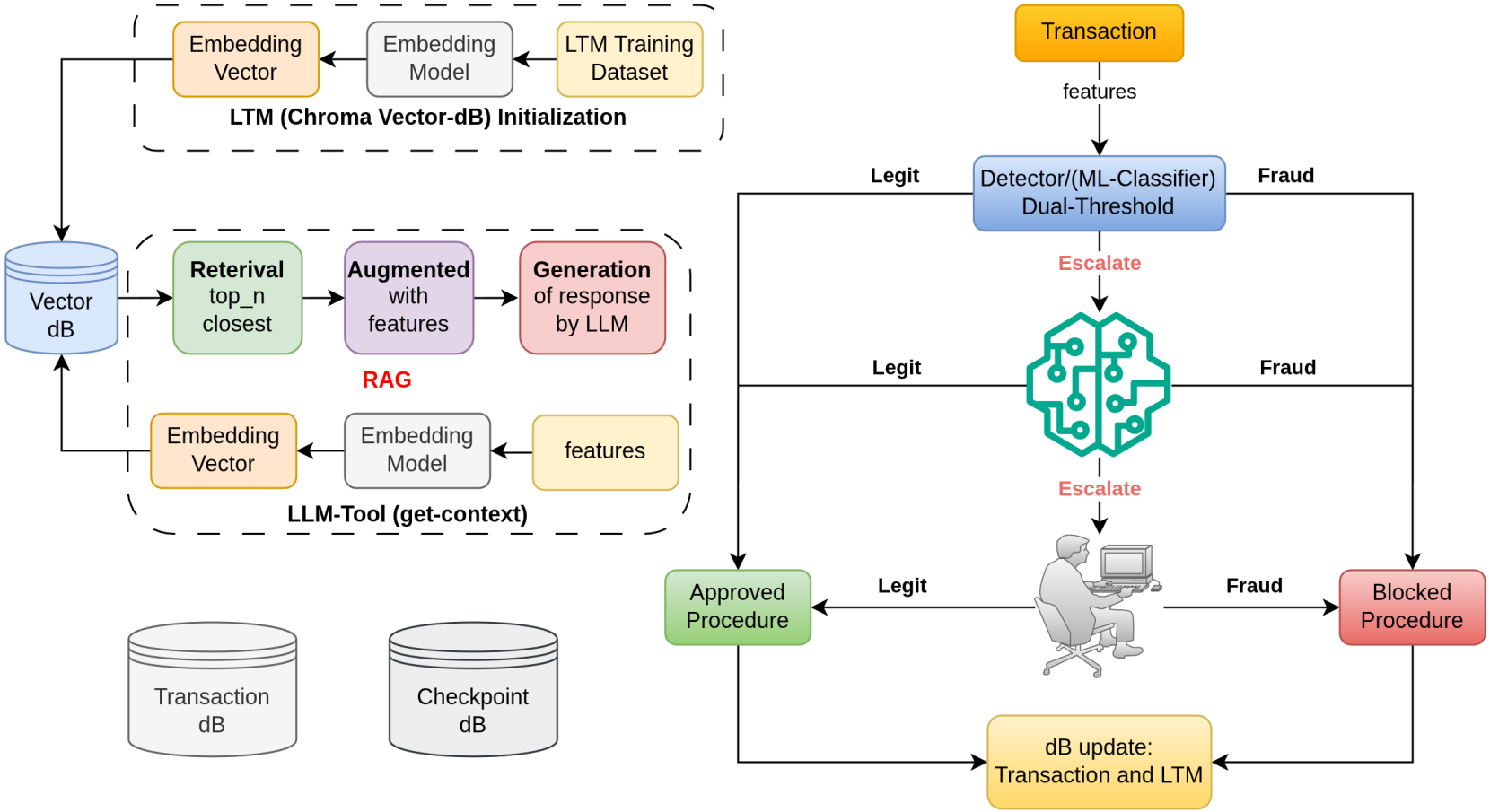
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1. Industry Context and Problem Definition

- Credit card fraud detection
- Critical & challenge in the **Financial Industry**.
- Heavily imbalanced (Fraud-Class < 0.2%)
- FN: cost, reputational damage
- FP: customer friction and operational review costs
- Balancing High Recall with Precision in automated risk mitigation
- AI can help

2. Integrated AI System Overview



3. Integration of Prior Capstone Projects

Capstone-3: Applied Machine Learning (Credit Card Fraud Detection)
Used XGBoost with cost-sensitive learning, and NO Over-Sampling
Here, Improved Feature Engineering of Time to 2-hour as hour-block
Here we use 4 different Over-Sampling (improving Recall)

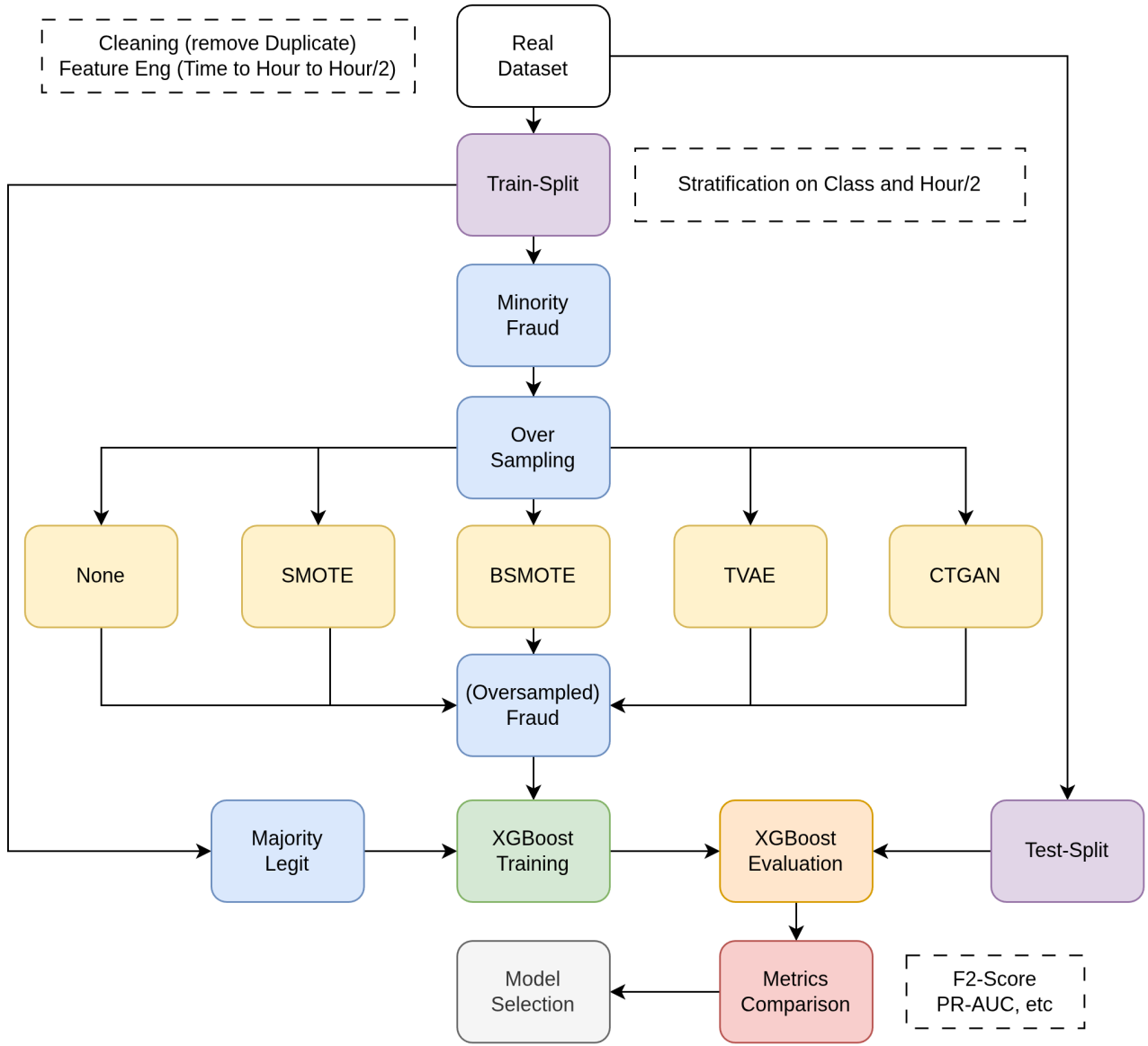
Capstone-5: Generative AI Applications (Anime Face Generation with VAE)
In Capstone-5 we developed VAE from scratch for image generation
Here we use Tabular version of VAE (TVAE) for transaction data generation
We also use CTGAN for data generation to select the best between TVAE/CTGAN

Capstone-6: Agentic AI Systems(Industrial Equipment Anomaly & Incident Response Agent)
In Capstone-6 No Real Dataset (data was simulated), no ML-Classifier, and No Vector-dB
Here We have Real-Dataset, Also we use Generative Model to generate new data
Hear We Use ML-Classifier with Optimized Dual Threshold as First Stage Detector
Also We have Vector-dB as Long-Term Memory for historical Context reterival (RAG)

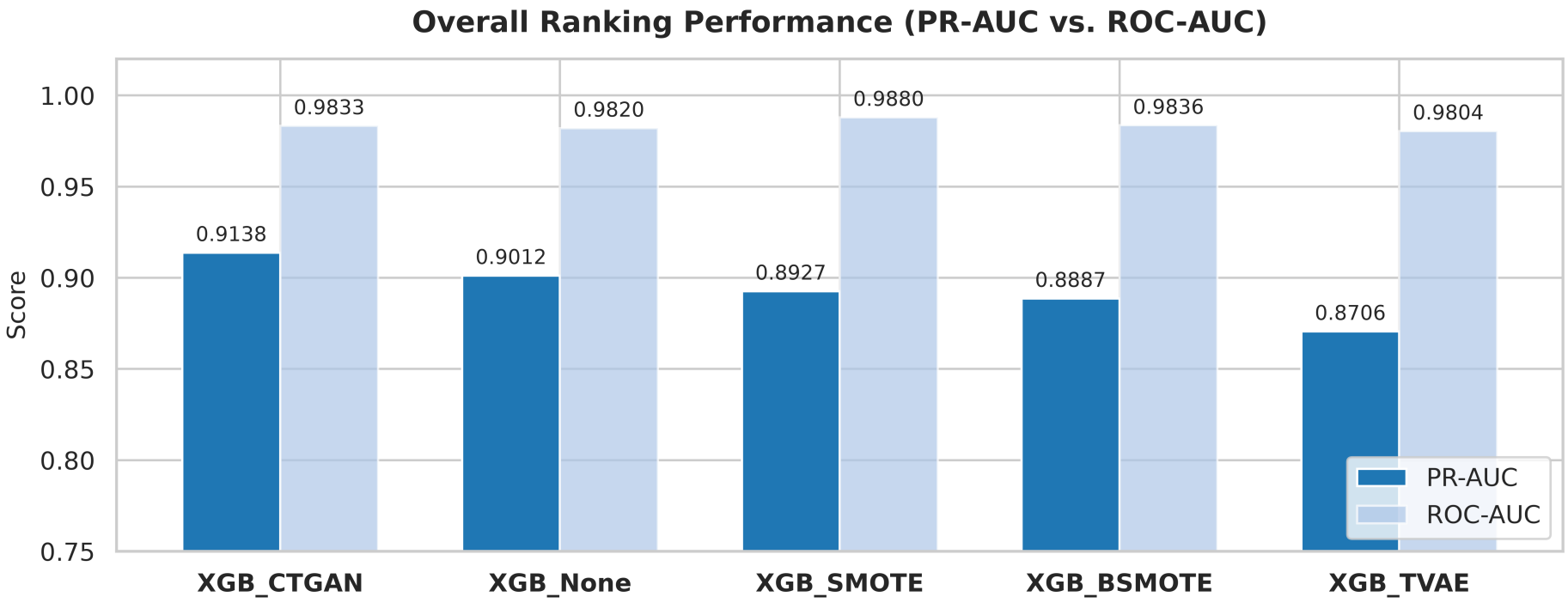
4. Key Technical Decisions and Tradeoffs

1. Step-1: Train, and Select ML Model
   (among 4 different Oversampling or No Oversampling)
2. Step-2: Train, and Select Generative Model
   (between TVAE and CTGAN)
3. Step-3: Define and Optimize Dual Thresholds,
   and Prepare Data for LTM initialization (warm-up)

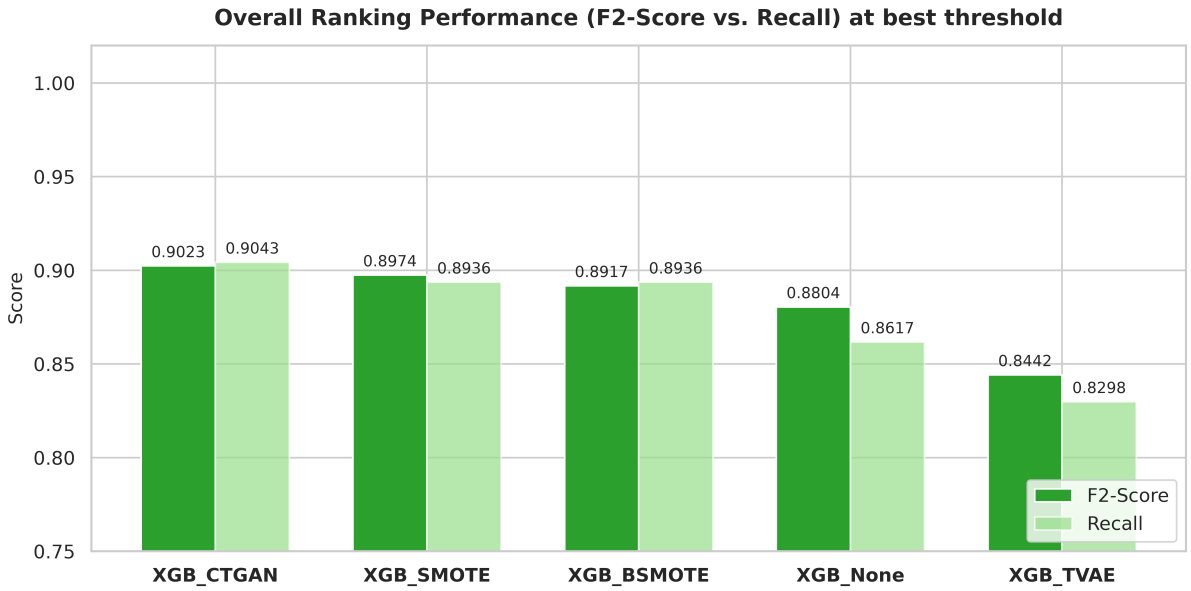
4. Key Technical Decisions and Tradeoffs (Cont. A)

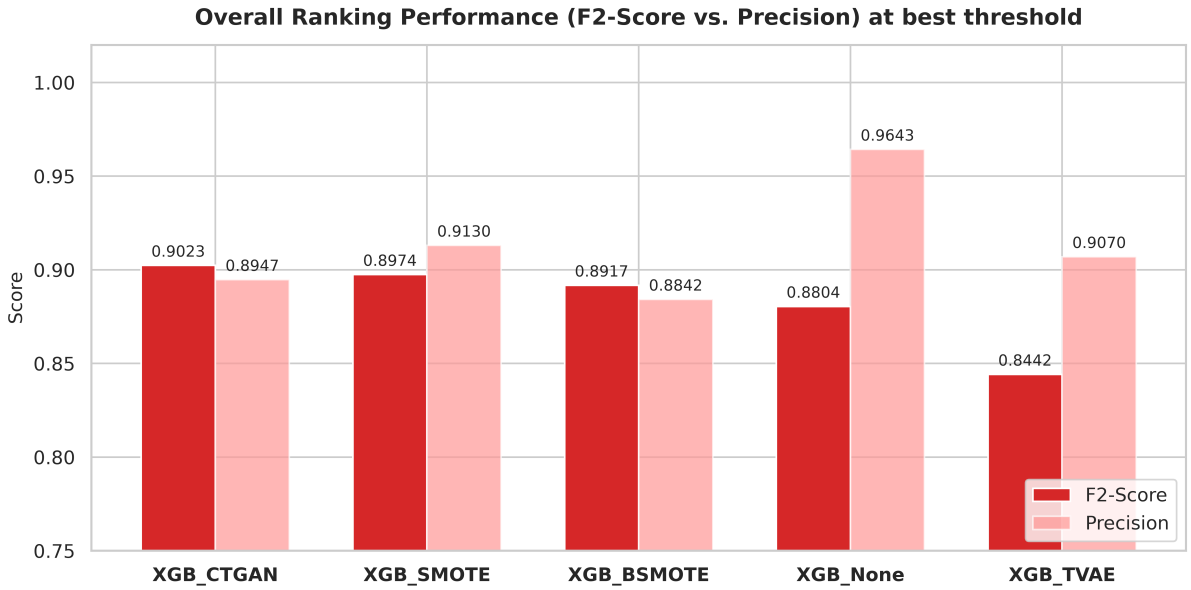


4. Key Technical Decisions and Tradeoffs (Cont. B)

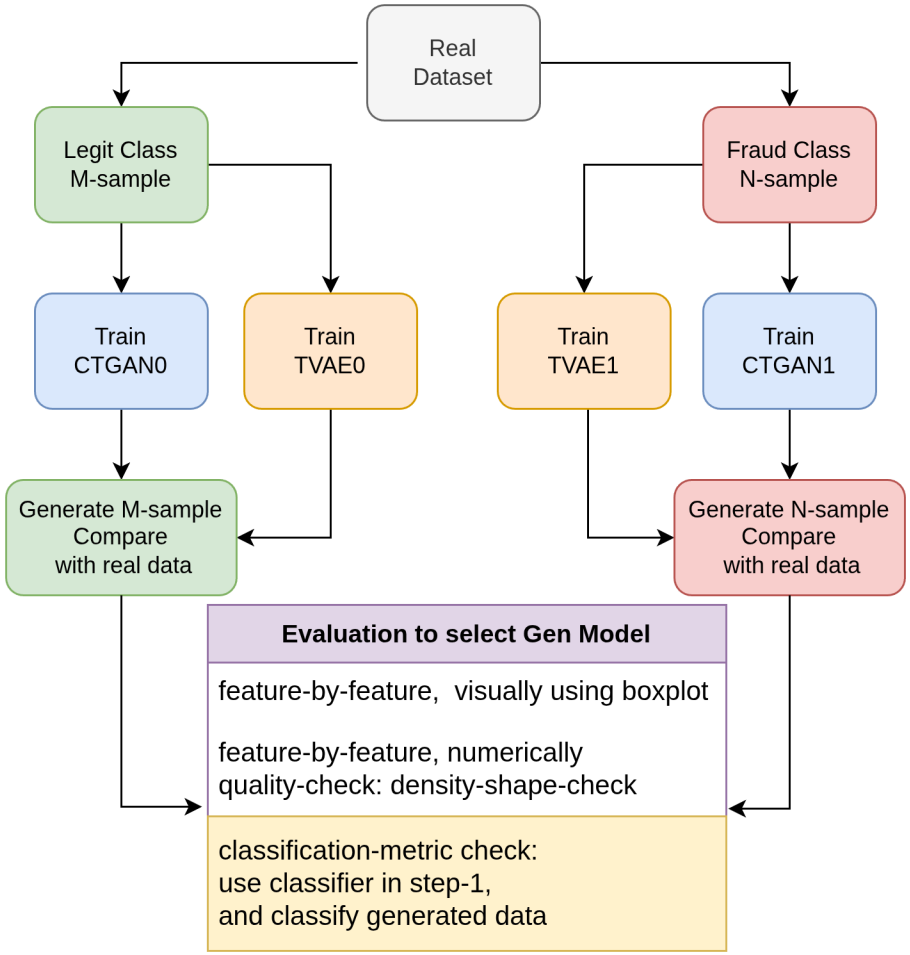


4. Key Technical Decisions and Tradeoffs (Cont. C)





4. Key Technical Decisions and Tradeoffs (Cont. D)

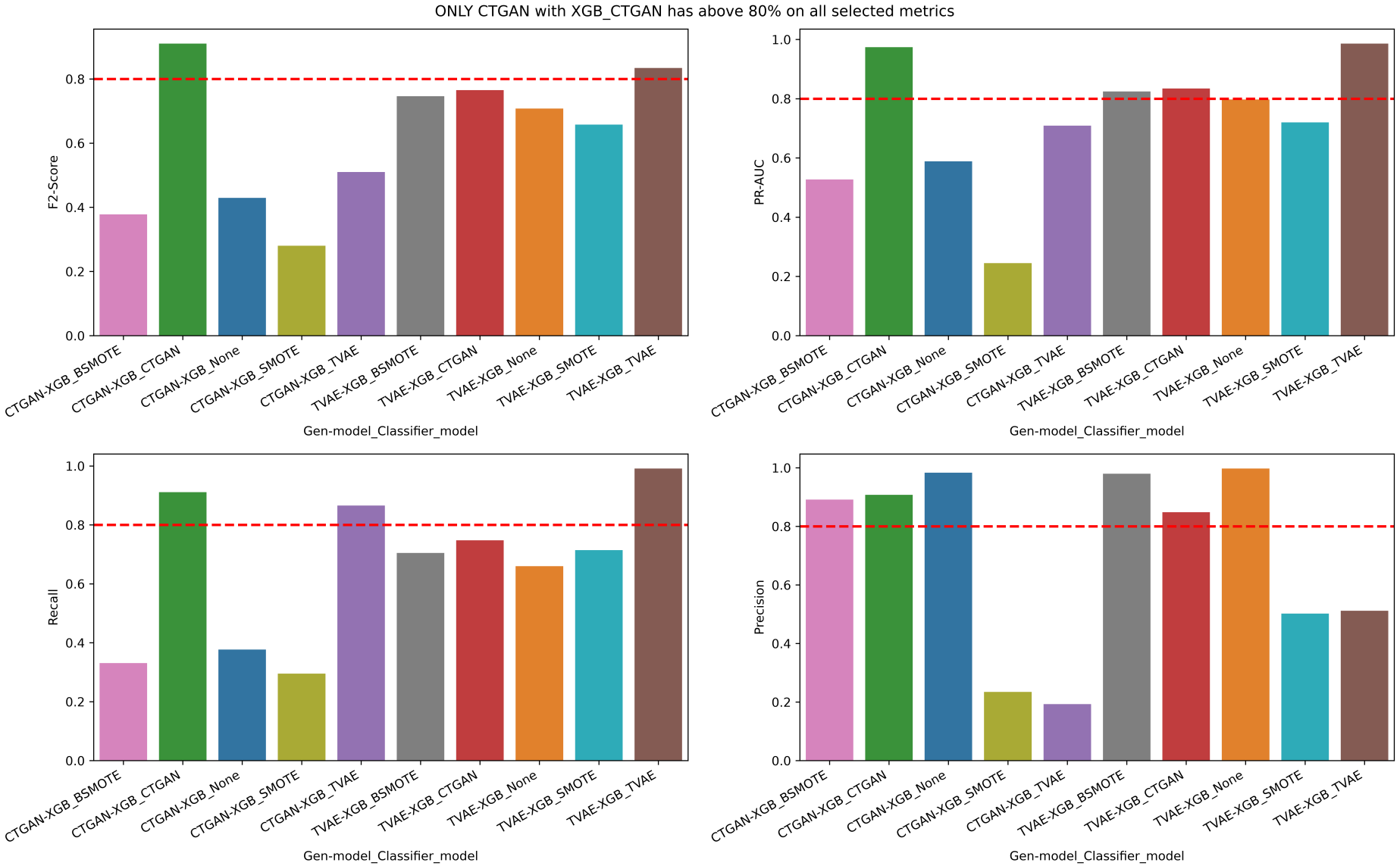


4. Key Technical Decisions and Tradeoffs (Cont. E)

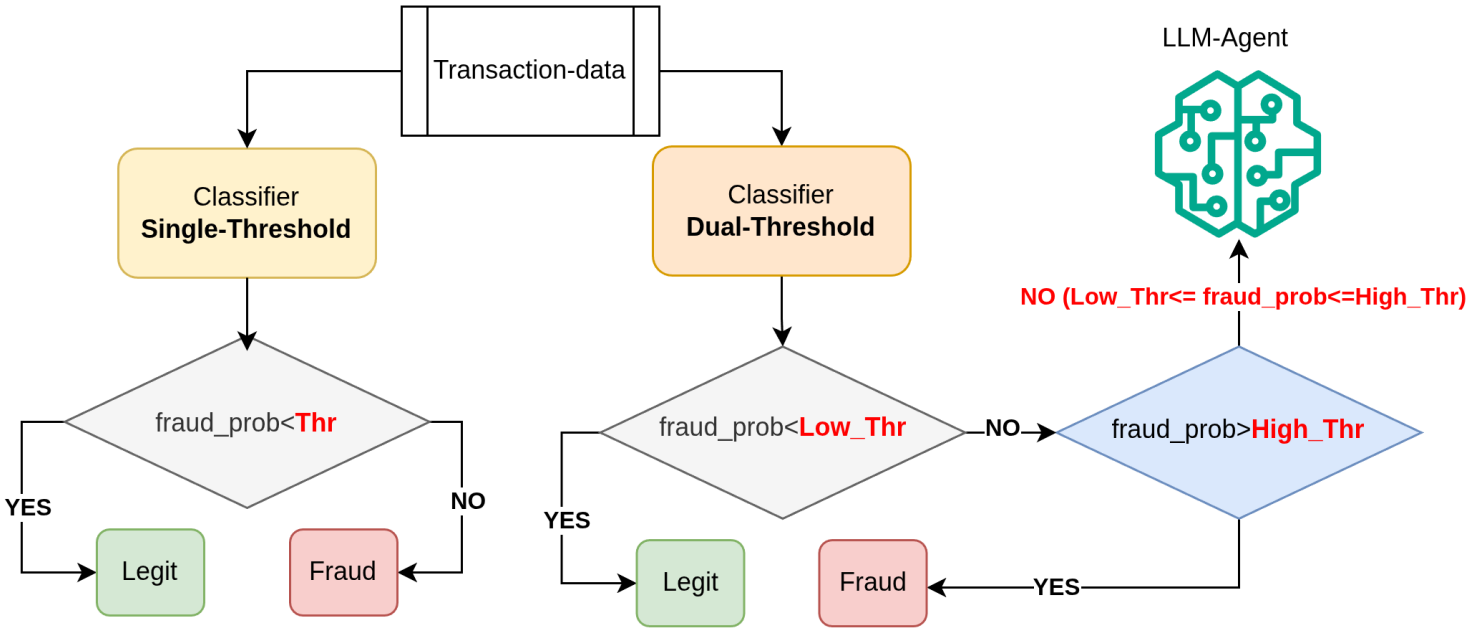
Table: Numerical Comparison of Generative models: TVAE vs CTGAN

Class Subset	Model	Overall Score	Feat. Wins (out of 30)
Fraud (*Class* = 1)	**TVAE**	**87.51%**	**24 (80%)**
	CTGAN	67.27%	6 (20%)
Legit (*Class* = 0)	TVAE	91.16%	13 (43.3%)
	CTGAN	**91.62%**	**17 (56.7%)**

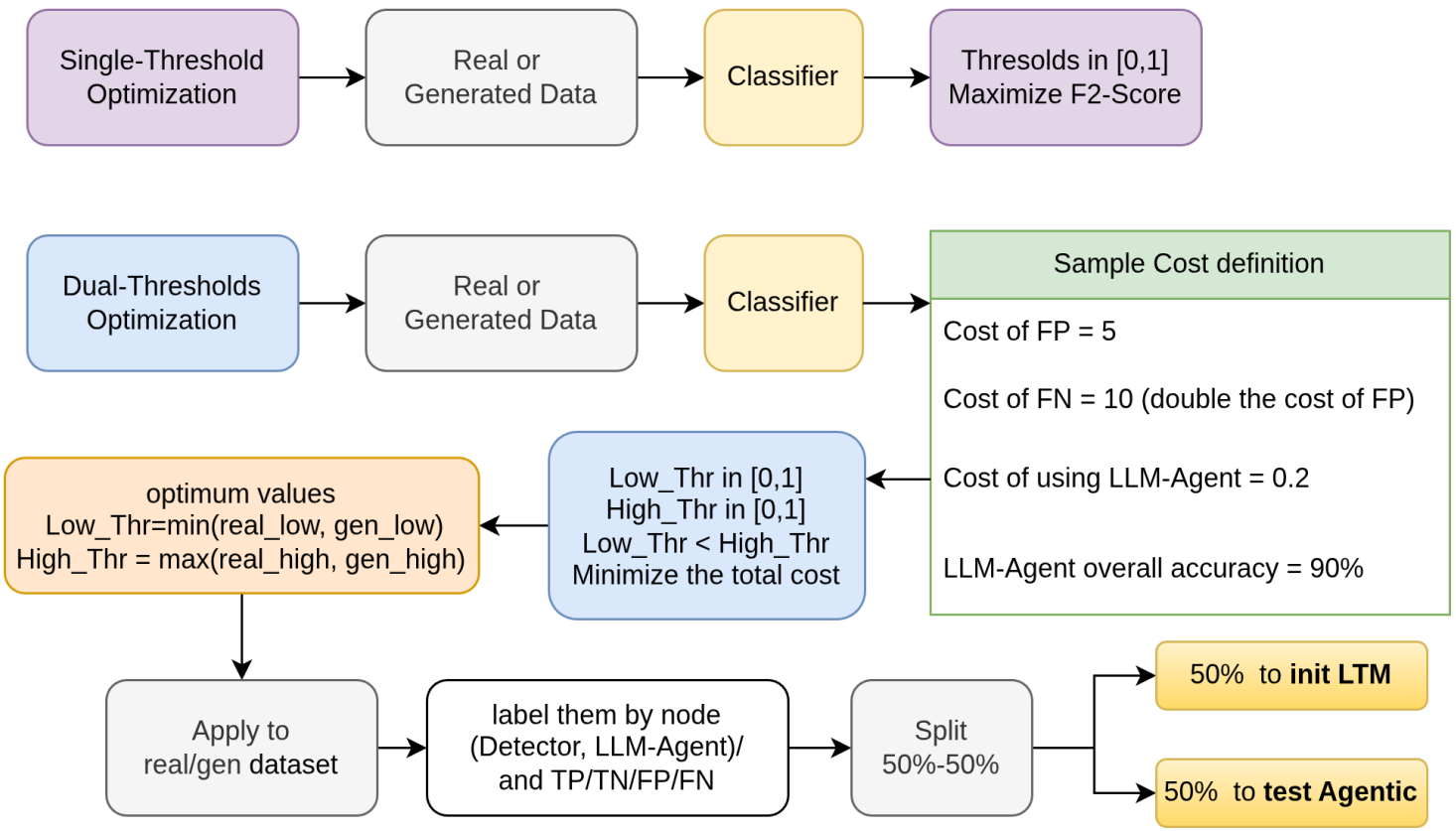
4. Key Technical Decisions and Tradeoffs (Cont. F)



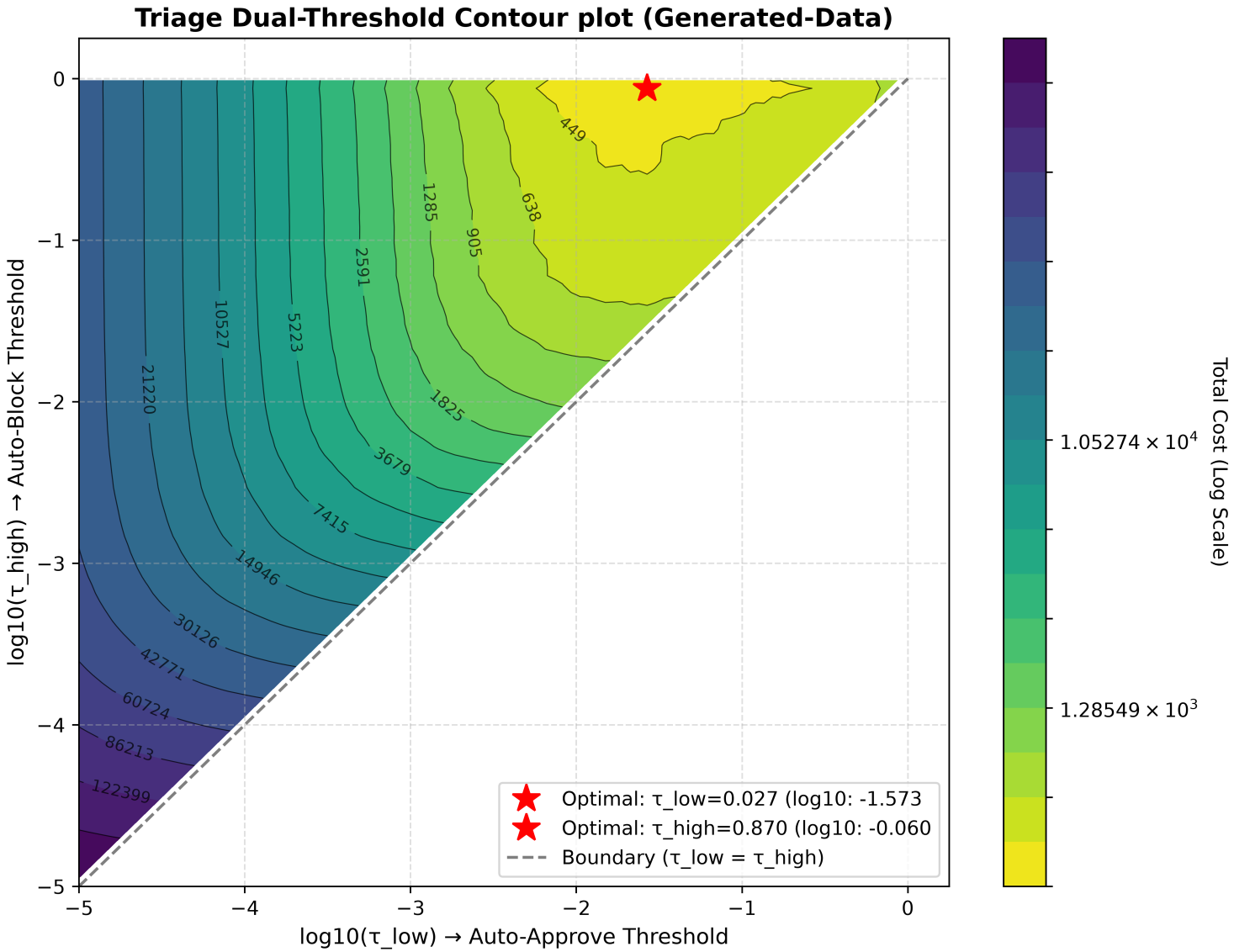
4. Key Technical Decisions and Tradeoffs (Cont. G)



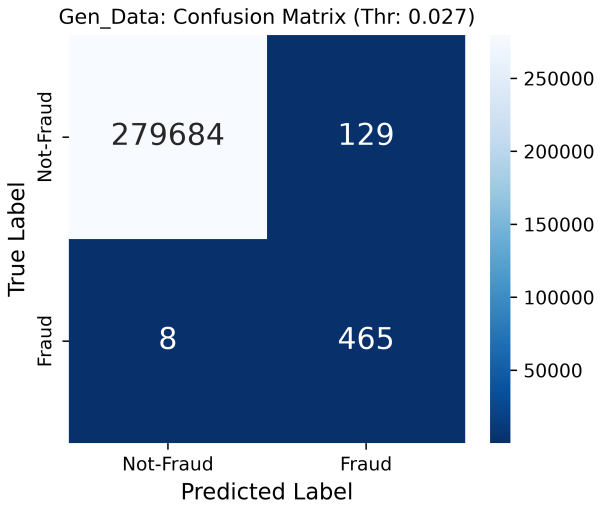
4. Key Technical Decisions and Tradeoffs (Cont. H)

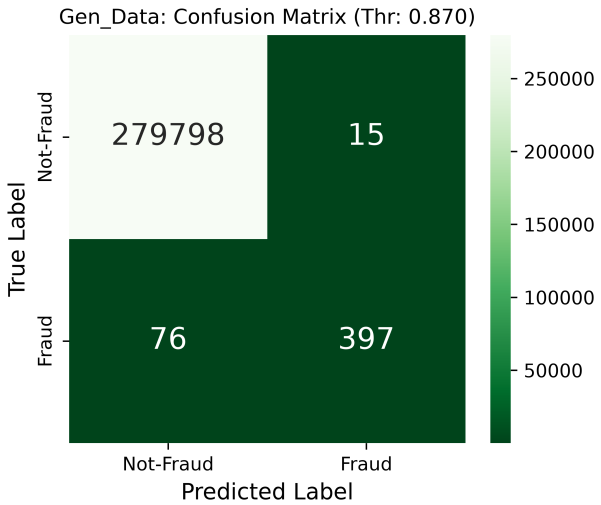


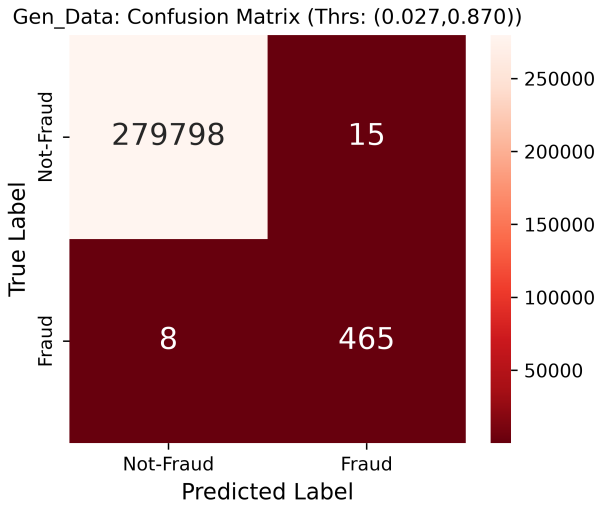
4. Key Technical Decisions and Tradeoffs (Cont. I)



4. Key Technical Decisions and Tradeoffs (Cont. J)







4. Key Technical Decisions and Tradeoffs (Cont. K)

1. Step-1: Train, and Select ML Mode: XGB_CTGAN
2. Step-2: Train, and Select Generative Model: CTGAN
3. Step-3:
   Dual Thresholds: Optimized,
   LTM-Warmup-Dataset: Prepared

5. Ethical Considerations and Responsible AI

- **Algorithmic Bias & False Positives:** I*mbalanced datasets risk generating biased rules that unfairly target specific demographic spending behaviors as anomalous.*
- **Human-in-the-Loop Accountability:** *By incorporating mandatory human escalation for low-confidence decisions, the system prevents autonomous wrongful account blocking and ensures user due process.*
- **Data Privacy:** *Synthetic data generation (CTGAN) helps mitigate privacy leakage by supplementing training pipelines without over-relying on sensitive real-world cardholder records.*

6. Evaluation, Limitations, and Risks

```
max@m223:~/x/zc07-ml-genai-agentic
max@m223:~/x/zc07-ml-genai-agentic$ uv run agentic_system.py --source real --node llm --label tn --test 10

[Check-dB] Transaction
[Check-dB] LTM-Chroma-Vector-dB
[Check-dB] Checkpoint
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Get/Gen] Transaction Samples
[Running] Test-id:10, Using real data with label:TRUE_NEGATIVE, and target-node:LLM_Agent
++++++++++++++++++++++++++++++++++++ 001/001 ++++++++++++++++++++++++++++++++++++
[TRANSACTION] V1:-6.3254, ..., V14:-2.0639, ...,  V28:-0.7889, , Amount:508.00, , Hour/2:6
[Detector-node] ESCALATE to LLM_Agent for extra investigation (fraud-prob: 35.34764%)
[LLM_Agent-node] ESCALATE to Human_Agent for extra investigation (fraud-prob: 35.34764%)
------------------------------------------------------------
[context] :
[doc-1]-id: 4bbefa11db, distance: 0.056398, Amount: 662.0, Hour_Block: 7, fraud_prob: 0.990794, actual_class: Legit
[doc-2]-id: 021296471f, distance: 0.066327, Amount: 319.2, Hour_Block: 8, fraud_prob: 0.996766, actual_class: Fraud
[doc-3]-id: 001f1b2599, distance: 0.122861, Amount: 195.66, Hour_Block: 7, fraud_prob: 0.999131, actual_class: Fraud

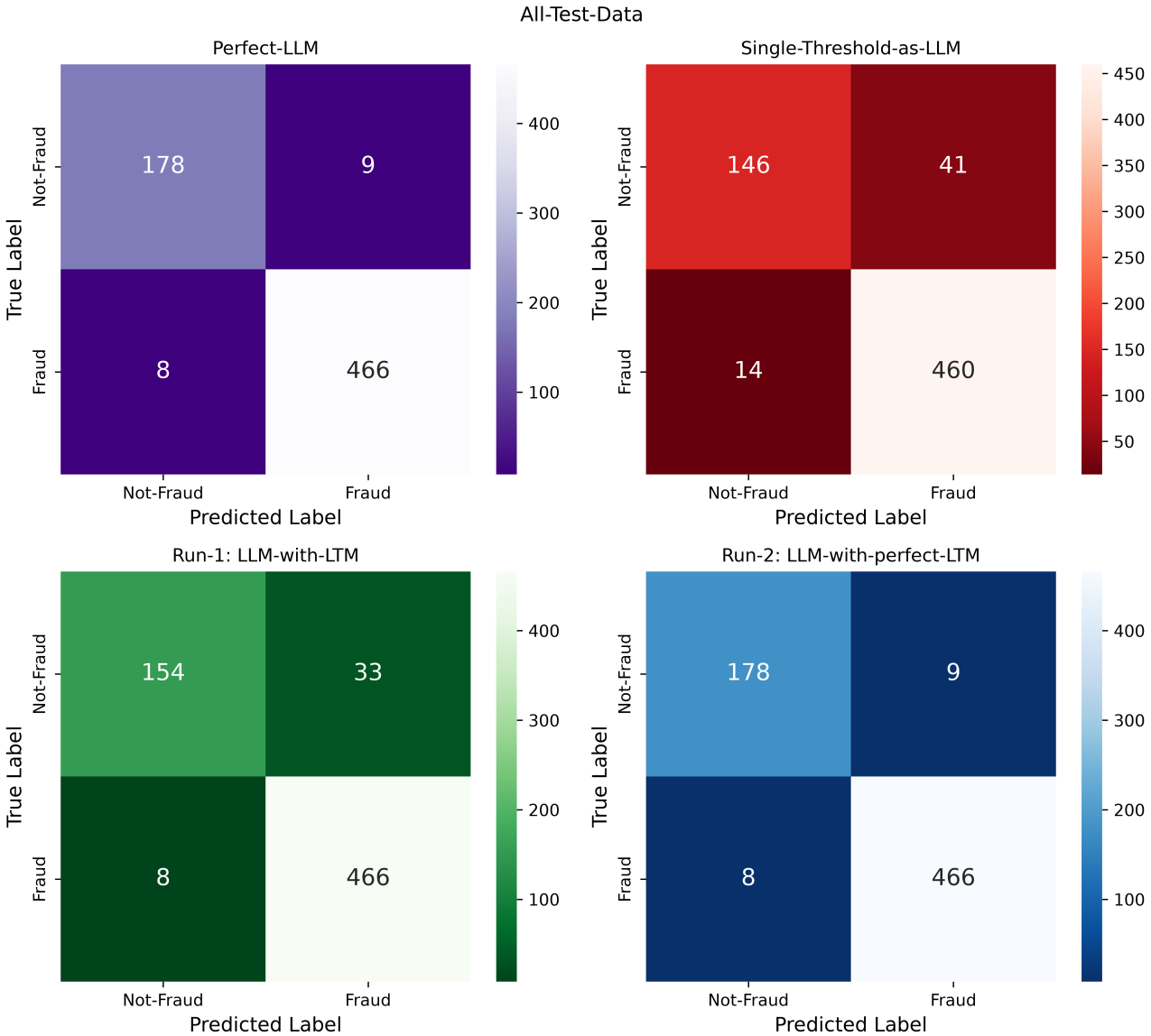
[llm_reasoning] The closest match ([doc-1]) is classified as Legit with a distance of 0.056398, while other matches include both
Fraud and Legit classifications. This creates uncertainty in the classification, necessitating escalation.
[llm_confidence] 0.5
[Human_Agent-node] Legit-Transaction send to Approved_Procedure (fraud-prob: 35.34764%)
[Approved_Procedure-node] Legit-Transaction APPROVED (fraud-prob: 35.34764%)
[Update_dB-node] 2026-10-10 13:22:43:transaction[6] logged to dB.TRUE_NEGATIVE NOT added to Vector-dB (LTM).
max@m223:~/x/zc07-ml-genai-agentic$
```

6. Evaluation, Limitations, and Risks (Cont. A)

Table: Test-cases Information.

node_source	TP	TN	FN	FP	Row_Count
Detector-real	225	50	4	2	281
LLM_Agent-real	8	21	0	0	29
Detector-gen	199	50	4	7	260
LLM_Agent-gen	34	57	0	0	91
Column_Sum	466	178	8	9	**661**

6. Evaluation, Limitations, and Risks (Cont. B)



6. Evaluation, Limitations, and Risks (Cont. C)

Table: Metrics for All Test-Case Run.

Run-Info	Accuracy	Precision	Recall	F1-Score	F2-Score
Perfect	97.43	98.11	98.31	98.21	98.27
Single-Thr	91.68	91.82	97.05	94.36	95.95
Run-1	93.80	93.39	98.31	95.79	**97.29**
Run-2	97.43	98.11	98.31	98.21	98.27

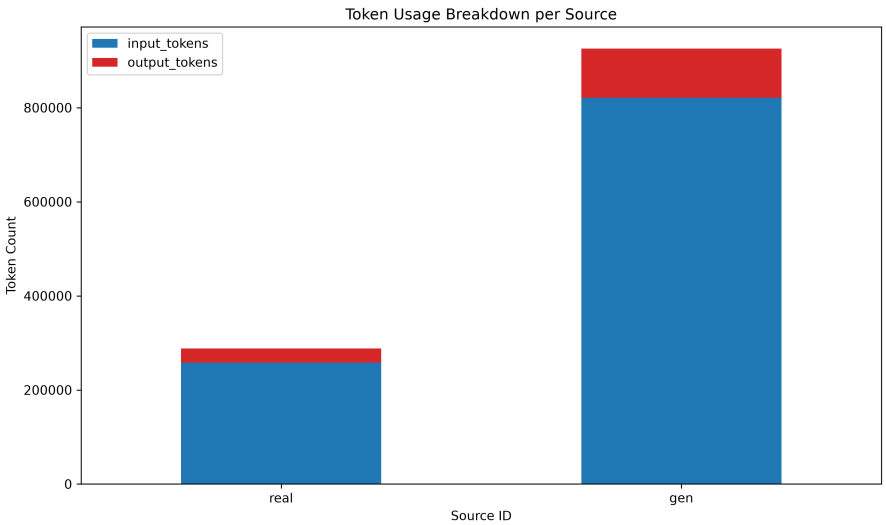
Based on the above result (Run-1), we can calculate agent-metrics as :

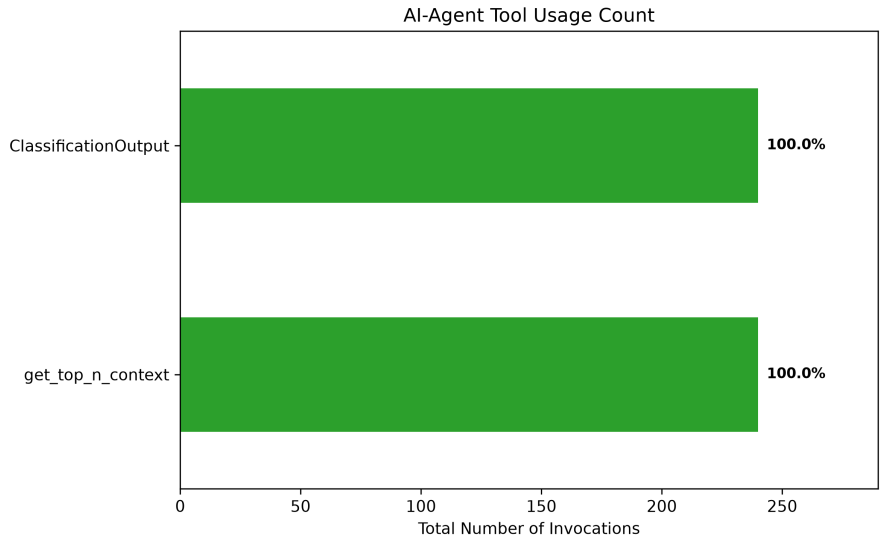
- **Accuracy**
  80.00%
- **Precision**
  63.64%
- **Recall**
  100.00%
- **F2-Score**
  89.74%

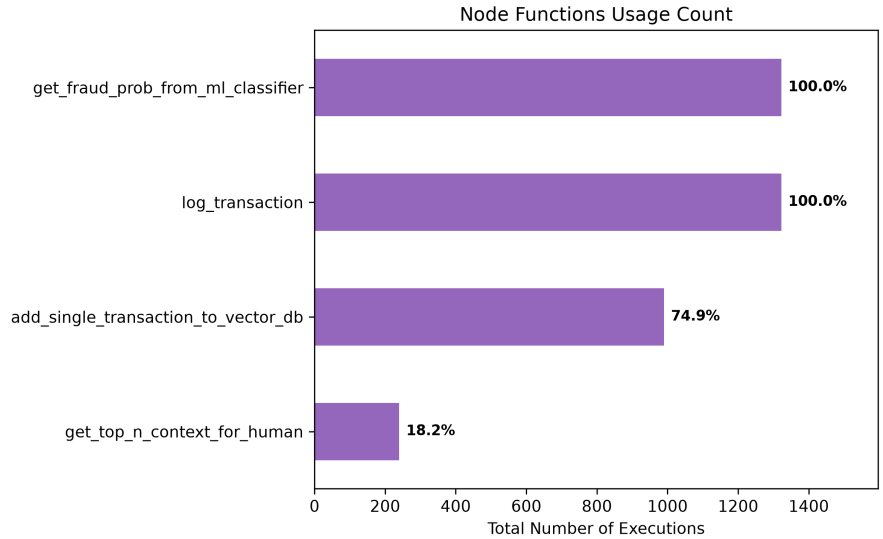
6. Evaluation, Limitations, and Risks (Cont. D)

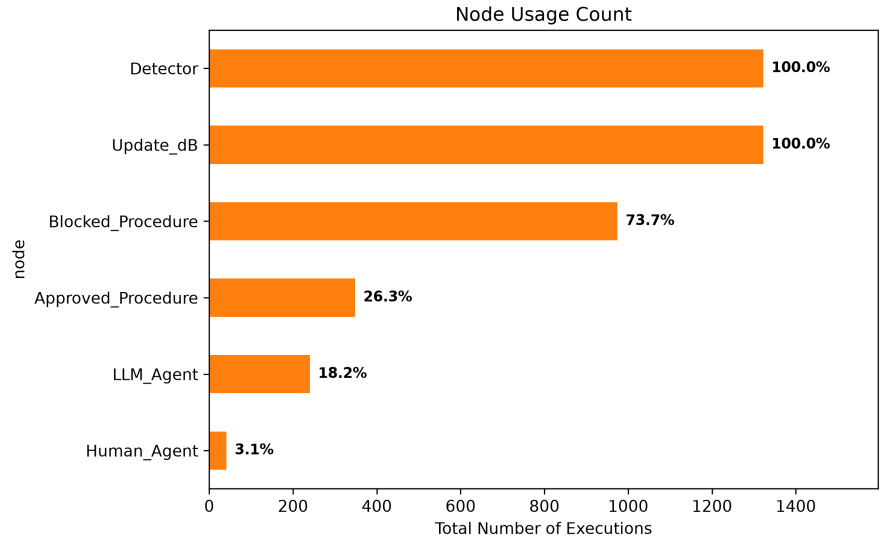
- **Successes:** Dual thresholds successfully filtered unambiguous traffic. LLM+LTM improved sequential accuracy, and human only involved when necessary.
- **Limitations:** Asymmetric precision/recall behavior (cause by Prompt) we need to decouple agent-specific precision and recall metrics during optimization.
- **Potential Metric Inflation**: Using same data in both step-1, and step-2 was necessary for initial feasibility, it can introduces *data leakage*, potentially leading to optimistic metrics. Let say, we have 5% performance inflation, then the realistic F2-Score is 92.2%.

6. Evaluation, Limitations, and Risks (Cont. E)









7. Professional Relevance and Next Steps

This project demonstrates end-to-end engineering readiness for modern AI roles by successfully bridging traditional predictive modeling, generative data augmentation, and agentic orchestration frameworks (LangGraph). It highlights a practical capability to design cost-aware enterprise systems that balance automated scale with human-in-the-loop safety.

7. Professional Relevance and Next Steps (Cont. A)

1. Refine the dual-threshold cost function (different precision and recall for agent)
2. Deploy real-time feedback loops to continuously update the vector database LTM
3. Expand automated evaluation suites to test resilience against evolving, adversarial fraud patterns.
4. To avoid possible data leakage due to usage of same data in step-1: ML-classifier and step-2 Generative Model, and test: When we have enough minority data we can split dat into 3 partitions. Also this split need to be done properly using stratification on continuous data which has its own challenge (we may need to use quantile-based discretization prior to stratification).